

3.2 – How to use the data

train_model 使用 PyTorch. 使用

****Train:**** 使用 CrossEntropyLoss — backward() Adam. clip_grad_norm_ RNN.

****Validation:**** 使用

Simple RNN, LSTM, GRU —

7 —

****** Accuracy, Val Loss F1-macro F1-macro — F1

****** Loss Accuracy train validation 15 train accuracy val

****Confusion Matrix**** fear sadness

GRU — val accuracy 91.2% — LSTM.

8 —

5 5

-
-
- fear sadness
- UNK
- love surprise

Figure 9 — Emotion prediction architecture diagram

The diagram illustrates the Emotion prediction architecture. It shows a sequence of input tokens being processed by an encoder (LSTM/GRU) to produce a hidden state. This state is then passed to a decoder (Simple RNN) which generates a sequence of output tokens. The output tokens are then passed to a softmax layer to predict the emotion.

The diagram shows the flow of data from the input text to the final emotion prediction. The input text is processed by the encoder, which produces a hidden state. This state is then passed to the decoder, which generates a sequence of output tokens. The output tokens are then passed to the softmax layer to predict the emotion.

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Emotion prediction architecture diagram

Q: What is the difference between LSTM/GRU and Simple RNN?

A: The main difference between LSTM/GRU and Simple RNN is the way they handle long-term dependencies. LSTM/GRU use gates to control the flow of information, allowing them to remember information for long periods of time. Simple RNN, on the other hand, only use a single layer of neurons, which makes them less effective at handling long-term dependencies.

Q: What is the purpose of the Embedding layer?

A: The Embedding layer is used to convert the input tokens into a numerical representation. Each token is assigned a unique vector, which is then passed to the encoder. The dimensionality of the embedding is typically 128.

Q: What is the purpose of the MAXLEN parameter?

A: The MAXLEN parameter is used to specify the maximum length of the input sequence. If the input sequence is longer than MAXLEN, the excess tokens are ignored. If the input sequence is shorter than MAXLEN, the remaining tokens are padded with zeros.

Q: What is the purpose of the padding parameter?

A: The padding parameter is used to specify the value used to pad the input sequence. In this case, the padding value is 0. This ensures that all input sequences have the same length, which is necessary for the model to process them.

Q: What is the difference between F1 and Accuracy?

A: Accuracy is a measure of the proportion of correct predictions out of the total number of predictions. F1 is a measure of the harmonic mean of precision and recall. F1 is generally considered to be a more robust measure of model performance than accuracy, especially in cases where the classes are imbalanced.